

project delivery. Availability of the DevOps specialist can be a challenge as the tools and technologies are vast in number.

- **Limited feedback channels:** A few enterprises follow traditional feedback channels like emails, which leads to delayed decision making and corrective actions.
- **Non-standard processes:** Teams sometimes adopt ad hoc implementation processes. There is an absence of centralised governance of approved practices.
- **End-to-end view missing:** End-to-end visibility of the engineering lifecycle in real time is missing, and there is an inability to govern custom implementation of pipelines in large enterprises.

DevOps implementation utilises automation tools covering continuous integration and delivery. Figure 1 depicts the DevOps adoption framework and the stages involved in DevOps implementation across enterprises.

DevOps readiness

This is the first step where the enterprise's readiness for DevOps adoption is assessed. Current DevOps capabilities and the tools usage across an enterprise are analysed to assess the DevOps adoption maturity level. Applications for DevOps adoption are identified based on the inputs provided by stakeholders.

DevOps planning and road mapping

DevOps planning involves the people, processes and tools to be used across an enterprise. The target stage is defined, and the transformation and execution plans are made. Business strategy and customer feedback are brought into the development lifecycle. DevOps objectives are finalised and an appropriate timeline for achieving them is estimated. A roadmap is developed to implement new DevOps processes, improve existing ones and make a list of the tools needed.

Baseline measurement: As part of baselining, the following data needs to be captured across the enterprise:

- Number of releases in a year
- Average time to fix a defect
- Time taken to complete the software build
- Time taken to complete execution of unit tests suite
- Average time to build and test

During this step, the DevOps domain architectures covering business, application, data and technology are defined.

DevOps solution blueprint

The solution blueprint addresses the implementation of continuous integration (CI), continuous testing (CT), continuous deployment (CD) and continuous change management (CCM) subsystems.

This blueprint supports both agile and non-agile development environments to work with physical, virtualised, and mixed hosts. It also supports configuration and control

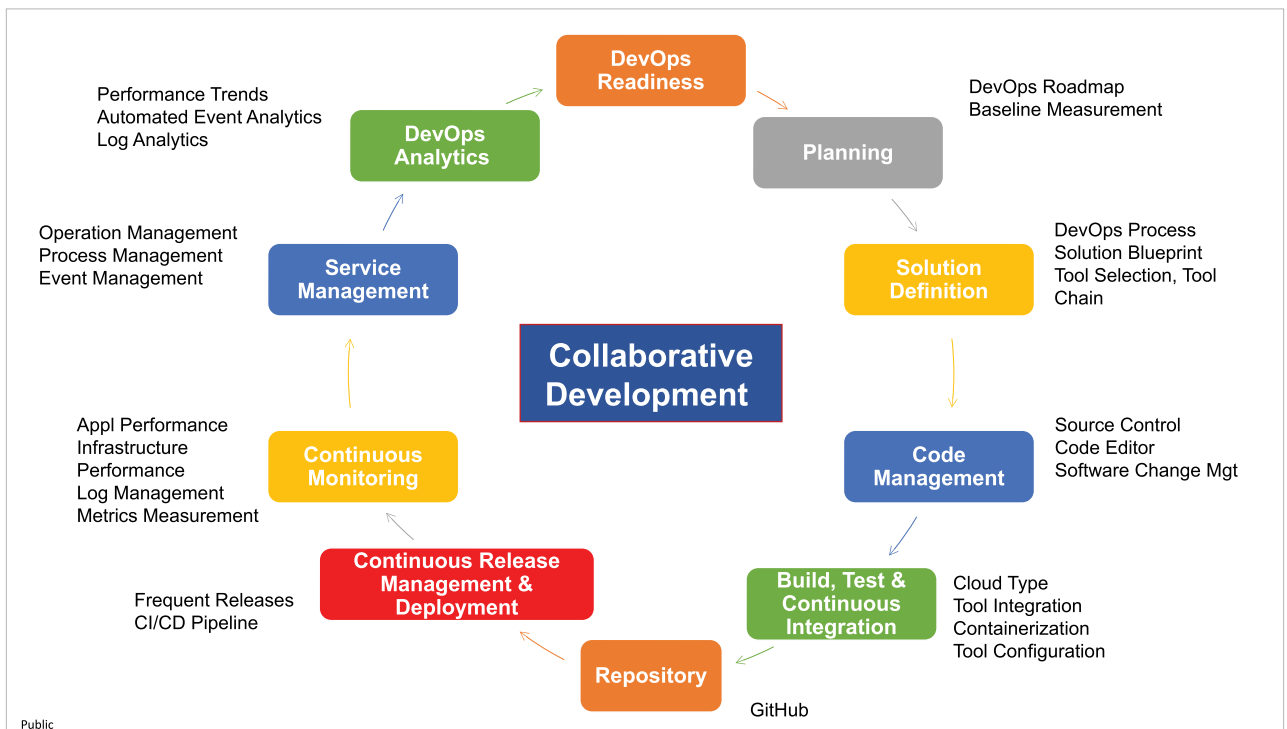


Figure 1: DevOps adoption framework